Analyzing the NHL Draft

CSE 163

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Background

Like many sports leagues, the NHL holds a draft each year in which each of the 32 NHL teams (previously fewer) select young players that they will have the rights to, given the player reaches the NHL level. Sports is a growing industry, and a lot of a team's success is determined by their ability to successfully scout players with the highest potential. In our project, we will examine how we can use statistics to draft players that we predict will have a higher success rate in the NHL. In our project, we analyze how different factors impact a players likelihood of being drafted into the NHL, and also impact their success in the league.

Data Settings

Given that we are combining three large datasets and we only want specific information from each, there was a lot of cleaning that needed to be done.Although data exists for years before 1998, we decided to work with only players drafted from 1998 to 2008 in order to be working with a manageable sized dataset. We also narrowed the skaters performance data to how they performed in their first 8 years in the NHL so that the results weren't skewed by certain players having played more years.

1. We imported the three CSV files to create three separate pandas DataFrames.
2. We noticed that the "pre_draft" dataset had multiple columns where the information was shifted and values were under an incorrect column name. In order to clean this dataset, we had to shift some columns over and delete others.
3. We narrowed down the data to only include players drafted 1998-2008. We also removed Goalies from all the data as their success is extremely difficult to compare to skaters.
4. The skaters NHL performance dataset has multiple rows for each player, representing their performance in each year they played in the NHL. We condensed this dataset so that each player was represented by a single row, summing certain values (like goals, assists, PlusMinus) and averaging others (time on ice per game, shooting percentage, shifts per game).
5. We made the player name the index for each dataset to simplify combining them.
6. The NHL draft dataset with each players "point_shares" also included multiple rows for each player so we averaged each players point shares over their years into one row.
7. We combined these three data sets by index, dropping unnecessary columns, and fixed up any small errors we noticed with the final dataset. For example, we removed a small amount of players where the position was entered as a letter that didn't match the majority (C, L, R, D).

Research Questions

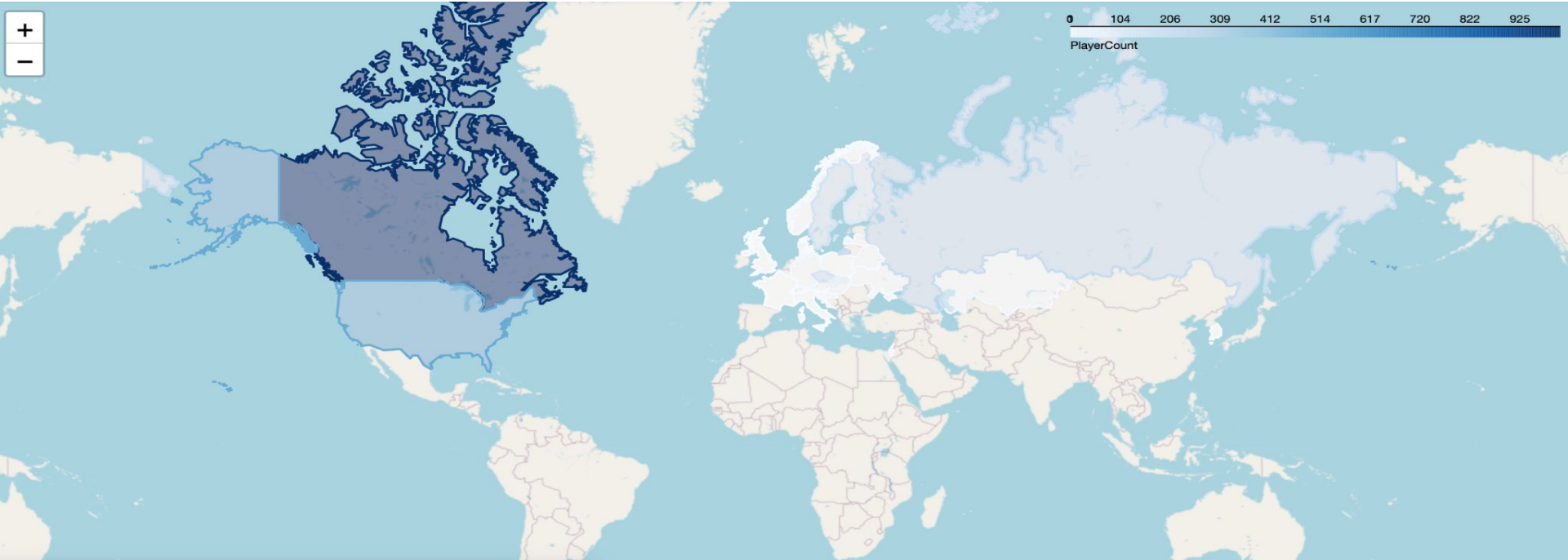
1. Which regions of the world are most represented in the NHL?
2. How does birth month influence ice hockey player draft selection order, considering the academic year cutoff?
3. How does a players ability to score in their junior league correlate with their success in the NHL?
4. How do the weight and height of a Player affect which Draft Round they are picked in?
5. Which positions are most valued for higher picks in the draft?

RESULTS

Interactive World Map

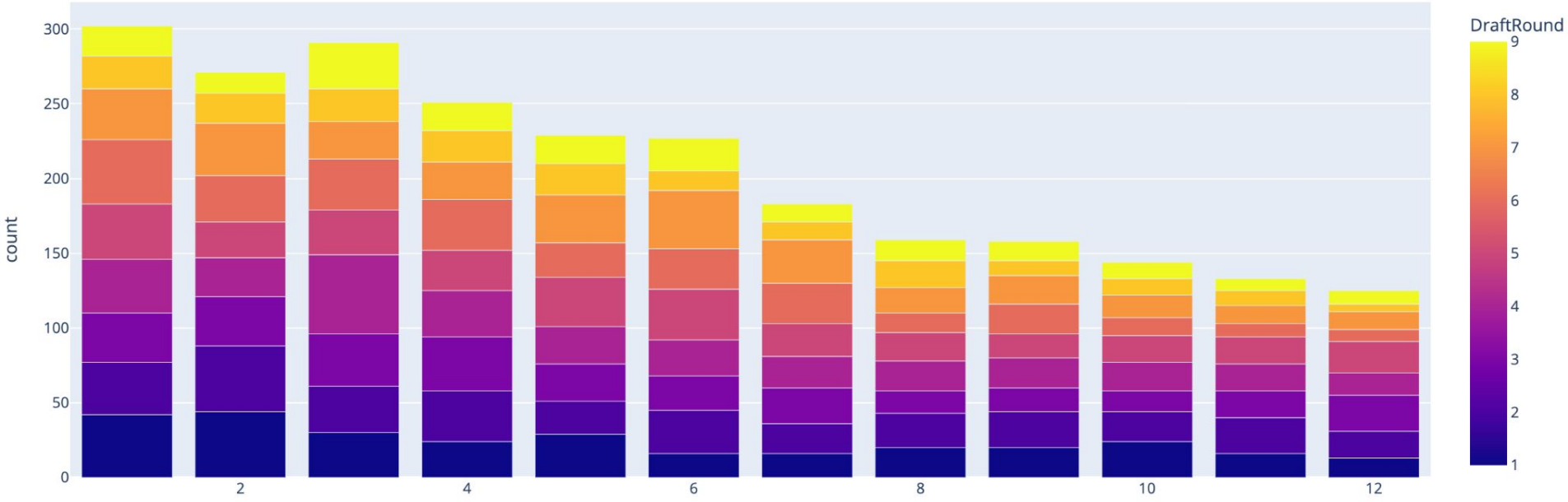
The majority of NHL players in our data set are from countries in the Northern Hemisphere, with Canada (1008 players), the USA (512 players), and Russia (238 players) being the top contributors. This trend is linked to the cold, wintery climates in these regions, which provide ideal conditions for the development of winter sports like ice hockey.

Given this information, NHL teams can send more scouts to countries where ice hockey is a further developed sport as they are likely to have more talented players.



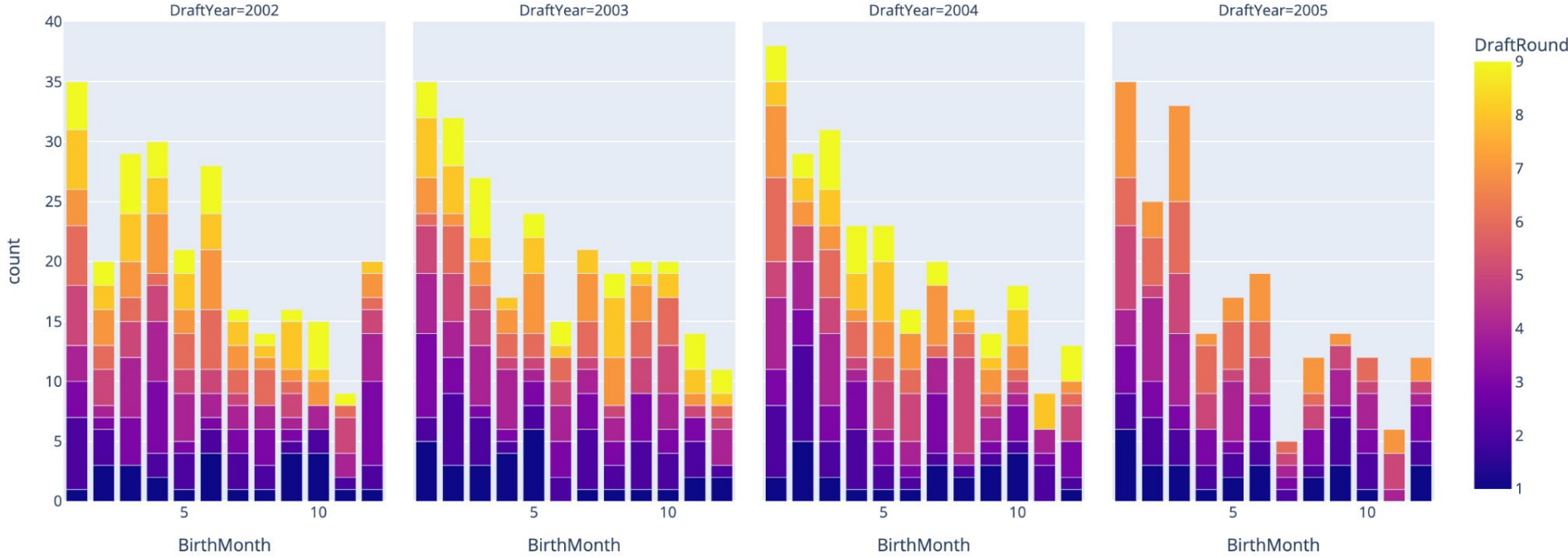
The distribution of all players from 1998~2008 by birth months

There is a decreasing distribution of players across birth months, from January and December, from the overall 8 years we are looking at in this project. In North America, youth leagues are organized by birth year (Jan-Dec). This trend suggests that players born earlier in the cutoff year have selection advantages, due to early physical maturity and greater opportunities in youth leagues. Earlier born players that are originally only slightly ahead in development, are selected for higher teams and receive better coaching, widening the skill gap between them and later born players even further.



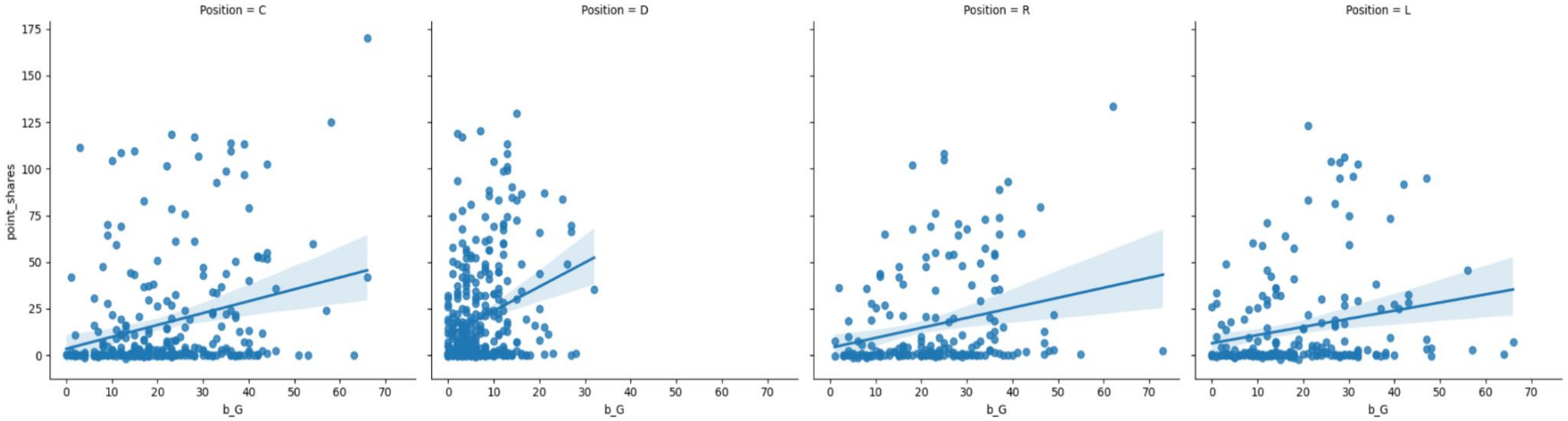
The Distribution of Players drafted from 2002 ~ 2005 by birth month

The decreasing trend is very pronounced in the years 2002, 2003, 2004, and 2005 draft rounds.



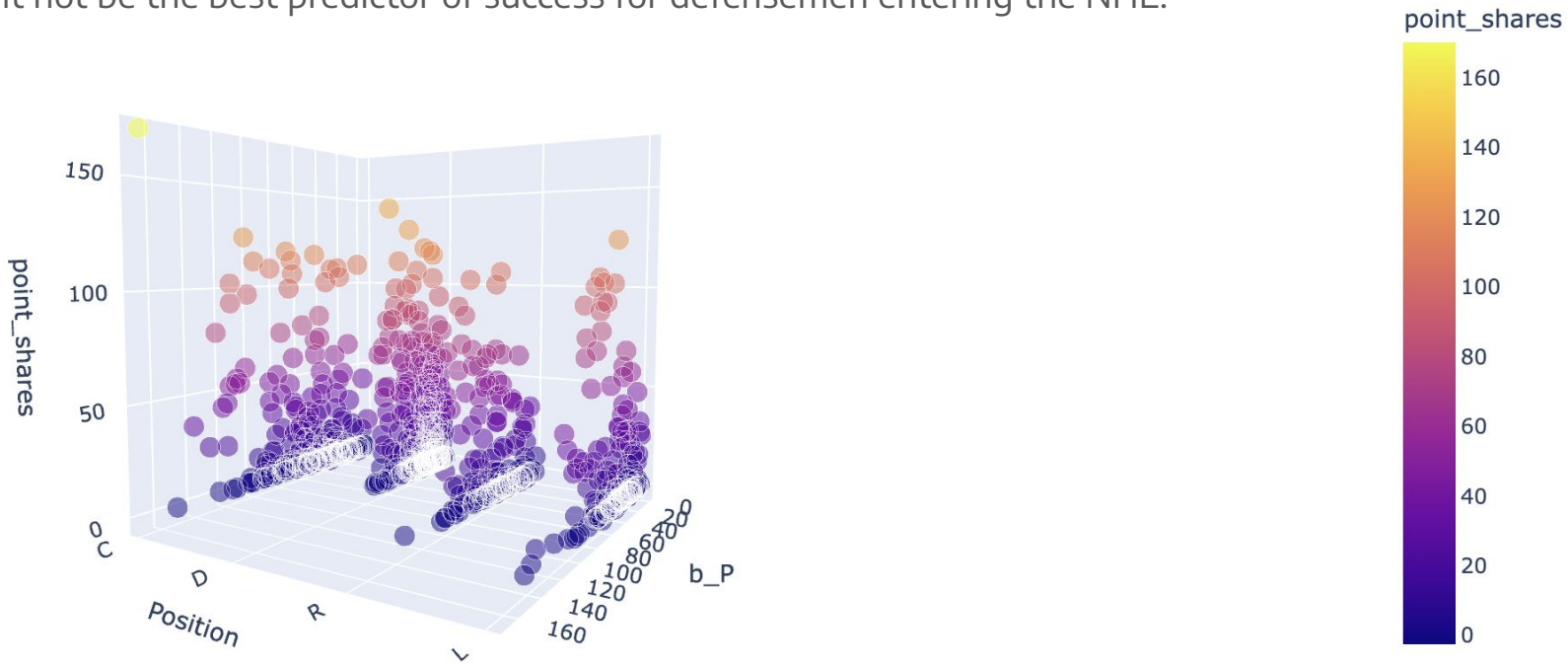
Do goals in junior leagues correlate to NHL success?

Point shares are a pre-calculated number given to each player in the NHL that we will use as a measure of success. We use this value over simply using points or PlusMinus because these values have varying levels of importance depending on a players position. For example, points might not be the most important identifier of success for a defender, but they are for a forward position. These subplots categorized by position help to show how goals scored before contribute to point share values in the NHL. Although the defenders score less goals overall, this plot shows that defenders who do score a lot have more success in the NHL compared to defenders who don't score as much.

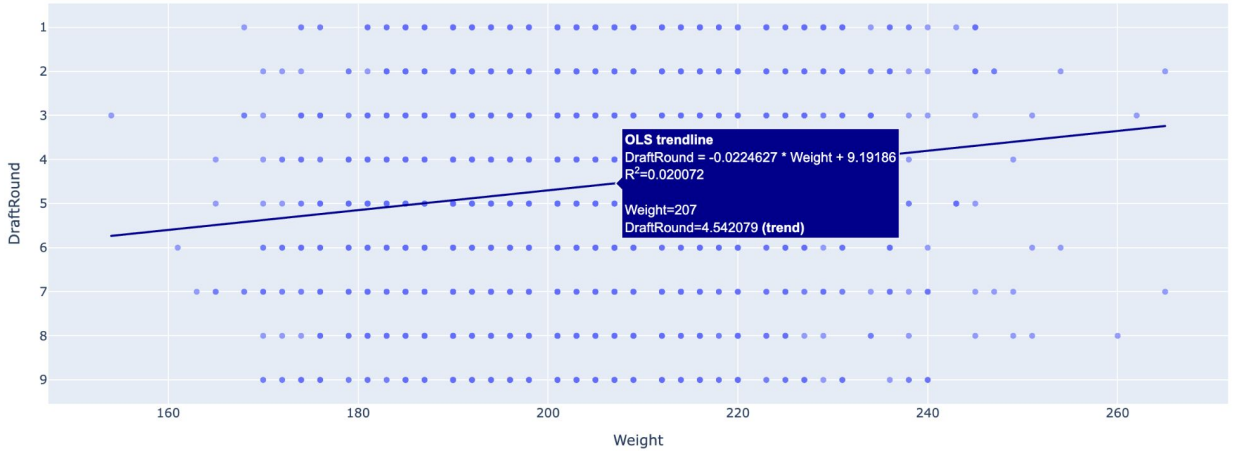


Success in NHL from offense in junior leagues

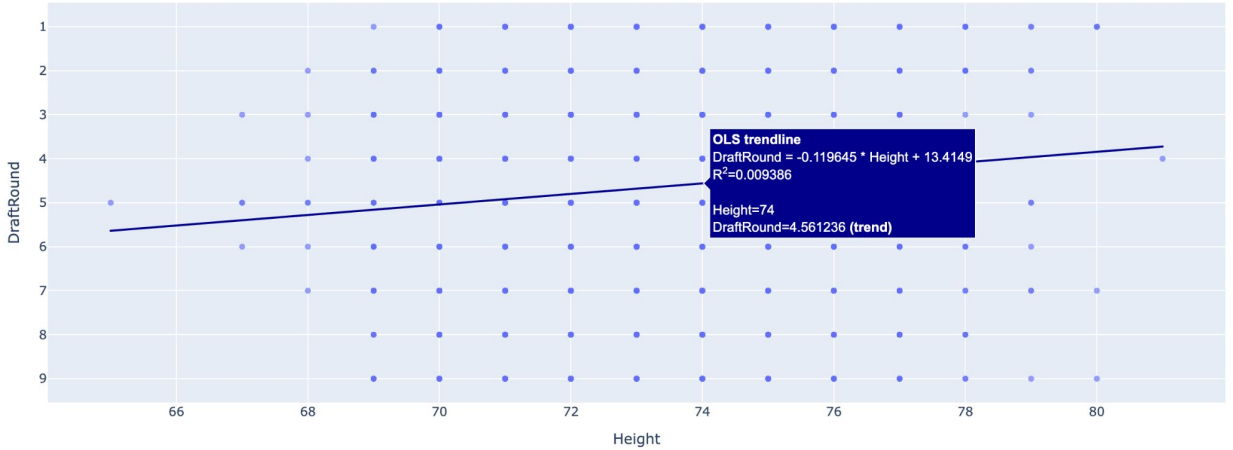
Because less offense (b_P or “before points”) is created by defense, but they are still able to have the same success as forwards in the NHL (point_shares), this 3-D graph shows that offensive statistics might not be the best predictor of success for defensemen entering the NHL.



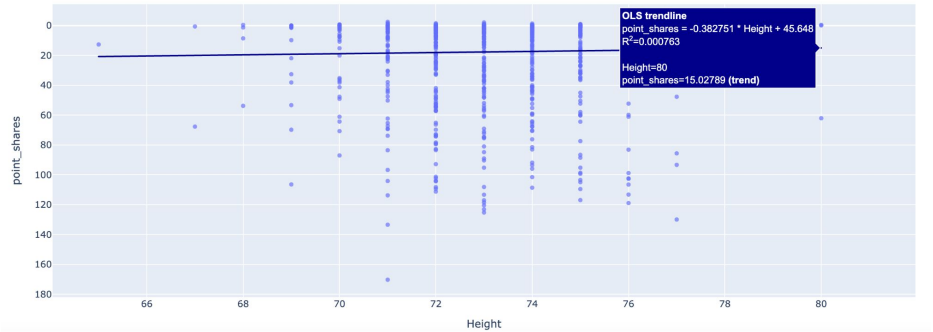
Weight vs Draft Round



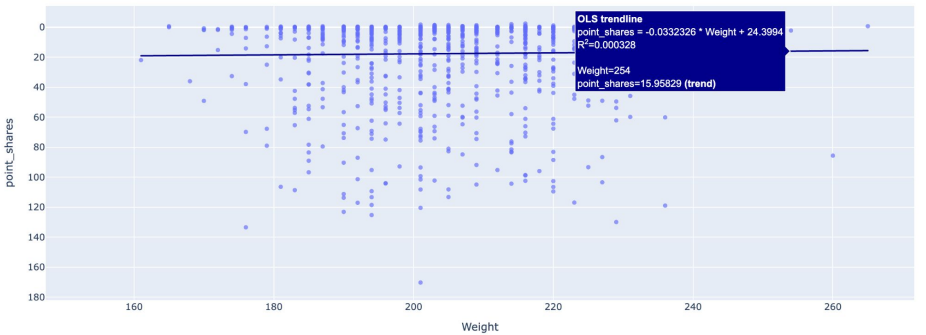
Height vs Draft Round



As seen in the previous slide, young players are more likely to get drafted, and in higher rounds, if they are further developed (higher weight and height). However, there is far less of a correlation between players height and weight and their actual success in the NHL. This suggests that height and weight are factors that NHL teams put excess value on when drafting players. They tend to draft players that are further developed at the time of the draft, when in reality, players that develop later end up succeeding just as much in the league.



Height and NHL Success



Weight and NHL Success



This pie chart shows the distribution of first round picks by position, excluding goalies. The distribution of players in the NHL is about 2/6 or 33% defenders, and 1/6 (17%) for each of the attacking positions. So why do centers get picked more often than the other positions. One possible reason for this is because centers are considered to make the biggest impact on a team offensively due to their size and ability to take face-offs.

Limitations

Our dataset may not account for all potential, sometimes unpredictable factors that influence draft success, such as injuries, coaching quality, or a players work ethic. We want to recognize that players' success is shaped beyond quantifiable metrics. Also, there might also be biases in the data because certain countries and leagues are often historically favored by the scouts, which may skew our results. Since we are only looking at the retired players from 1998 ~ 2008 to ensure fairness, this might pose a limitation to our analysis, because the NHL continues to globalize and teams may prioritize different scouting regions or place greater emphasis on new performance metrics. Another key limitation is distinguishing correlation from causation. We made observations such as the birth month correlates with the draft orders, but that doesn't mean one causes another.

Implications

Our project's analysis provides insights for NHL teams and scouts by identifying key factors that contribute to draft success, helping refine recruitment strategies with data-driven decision-making. It can also benefit players by highlighting trends that may improve their chances of being drafted. While there are limitations, such as potential biases and the exclusion of late-developing players, these findings serve as a strong foundation for further research and can complement qualitative scouting methods, leading to more informed and effective player evaluations.